

CodeVector Internship - Take home task

Expected time - 3-4 hours.

Submission time - within 3 days of receiving this task

What this is? and what we're really looking for

A small backend task. It's deliberately simple, the point isn't writing lots of code (AI can do that) it's making something correct and fast on real-sized data, and being able to explain why you built it that way. We're testing your software development skills, not your AI usage. This entirely remote role and you work in the founder's office, you'll own real, meaningful engineering challenges from day one. not mundane side tasks which can be automated with AI.

A note on using AI

- Use ChatGPT / Claude / Cursor etc freely. Using AI is allowed.
- In the next round you'll explain and change your own code live, so build something you actually understand. Or take time to learn and understand, We highly appreciate the learning attitude.
- If asked to build without AI and if you are not capable of coding it yourself, then stop. Please don't just paste it into AI and get a solution. AI is supposed to complement your skills and make implementation faster.

The task

Build a small backend that lets someone browse ~200,000 products (newest first), filter by category, and paginate through them.

1. Pagination should be fast.
2. Show the correct data while data is changing. If 50 new products are added/updated while someone is browsing, they must not see the same product twice or miss one.

We are intentionally not telling you exact requirements, working out the right approach, and learning it if it's new to you, is the whole point. Preferably Python or NodeJs. And Choose any DB you find fit.

The data

In the database, generate the products yourself. Write a small script that creates **200,000 products**, each with a name, category, a price, a unique id, updated_at, created_at time. It's fine for many products to share the same column values. Commit this generation/seed script to your repo so we can see how you did it. Tip: don't do a slow approach in a loop.

Bonus - optional

- A simple UI to browse the data — build it any way you like with AI, design it entirely with AI if you want. We don't read or grade UI code; this task is about the backend.
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Where to host (we encourage free, no credit card required platforms)

- **Backend:** Render or any online platform which has free web service.
- **Database:** Neon or Supabase or any free online database provider.

What to send us

- The live public hosted URL.
- Your GitHub repo. If private repo [invite us](#)
- **A short note:** what you chose and why, what you'd improve with more time, and how you used AI (what it helped with, and anything it got wrong that you caught).
- Bonus points for sharing your entire AI chat. We're genuinely interested in how you worked.
- Email to siddharth@codevector.in

Next and final round - 1 Hour online interview

- we go through the submission you made.
- We talk about what you learnt doing this.
- We ask you to make a small change to your own code, live.

Keep it small, make it correct, learn what you need, and be honest. Any questions? just ask us.